



NATIONAL UNIVERSITY OF MODERN LANGUGES

Information Security

Assignment : 1

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Assignment : Information Security

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Packet Analysis Report Using Wireshark

1. Introduction to Wireshark

Wireshark is an open-source network protocol analyzer used to capture and inspect network traffic in real time. It helps cybersecurity professionals, network administrators, and students analyze packets traveling through a network.

Wireshark allows users to:

- Capture live network traffic
- Filter specific protocols (HTTP, TCP, DNS, etc.)
- Inspect packet details
- Detect network problems
- Identify suspicious activity

2. Objective of the Experiment

The objective of this task is:

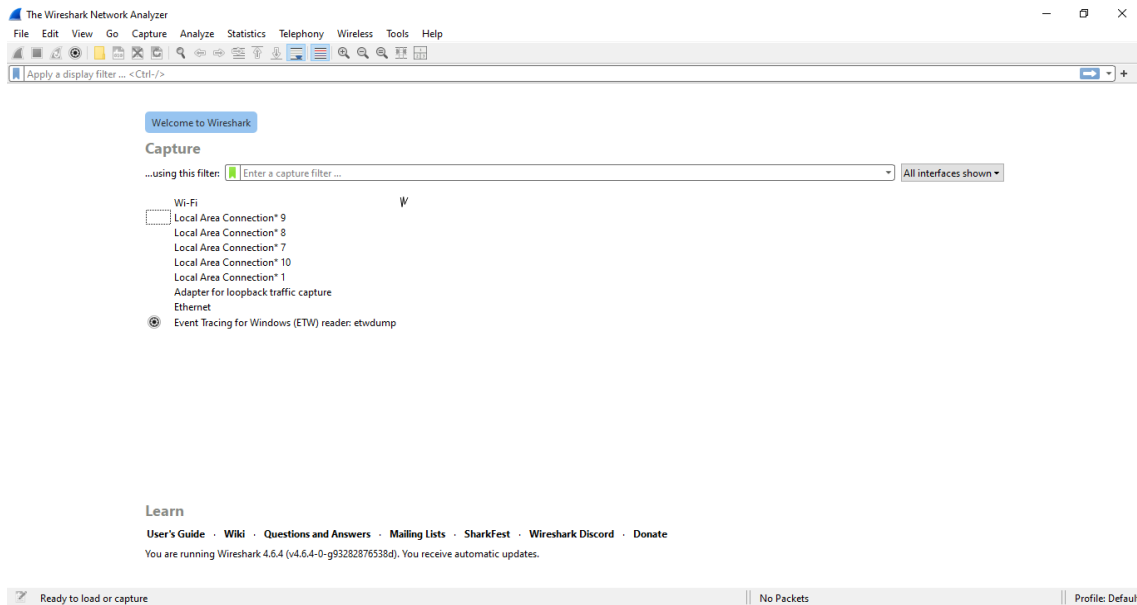
- To capture live network packets using Wireshark
- To analyze different protocols
- To understand packet structure
- To identify source and destination addresses
- To observe communication between client and server

3. Methodology

Step 1: Installing Wireshark

- Downloaded Wireshark from the official website
- Installed the software
- Opened the application

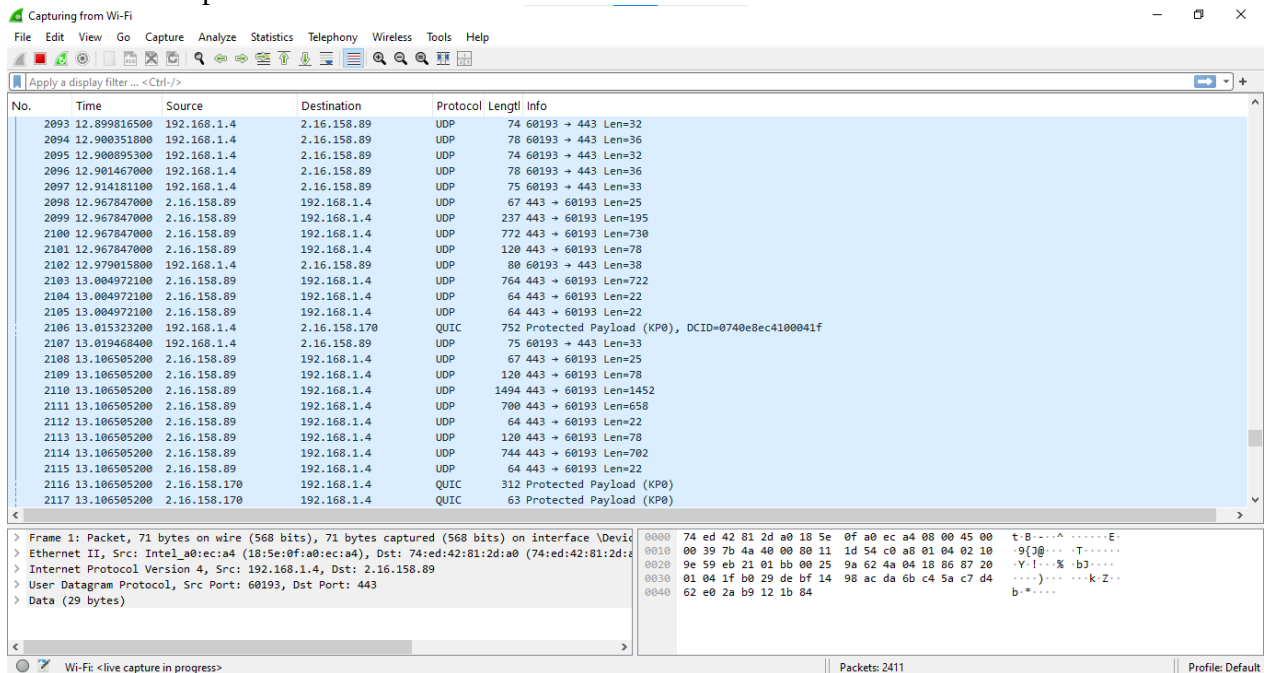
Wireshark Home Interface



Step 2: Starting Packet Capture

- Selected active network interface (Wi-Fi/Ethernet)
- Clicked on **Start Capture**
- Generated traffic by browsing websites or using ping command

Live Packet Capture



4. Packet Analysis

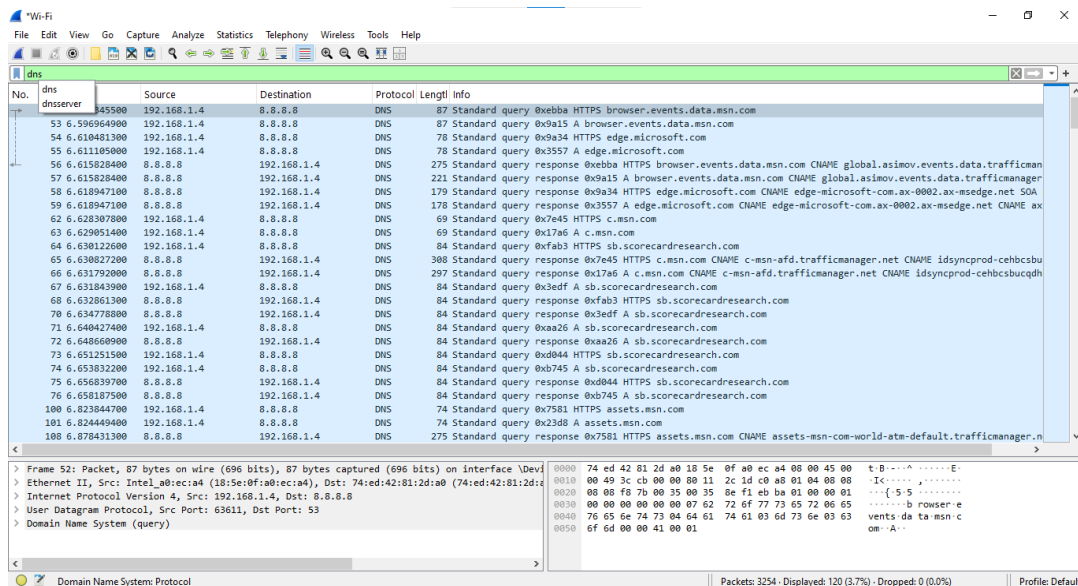
4.1 DNS Packet Analysis

Protocol: DNS

Purpose: Converts domain name into IP address

Observations:

- Source IP: (Your IP address)
- Destination IP: DNS Server
- Query Name: example.com
- Query Type: A (IPv4 address)



No.	Time	Source	Destination	Protocol	Length	Info
53	6.596964900	192.168.1.4	8.8.8.8	DNS	87	Standard query 0xebba HTTPS browser.events.data.msn.com
54	6.610481300	192.168.1.4	8.8.8.8	DNS	87	Standard query 0x9a15 A browser.events.data.msn.com
55	6.611105000	192.168.1.4	8.8.8.8	DNS	78	Standard query 0x9a34 HTTPS edge.microsoft.com
56	6.615828400	8.8.8.8	192.168.1.4	DNS	78	Standard query 0x3557 A edge.microsoft.com
57	6.615828400	8.8.8.8	192.168.1.4	DNS	275	Standard query response 0xebba HTTPS browser.events.data.msn.com CNAME global.asimov.events.data.trafficmanager.net
58	6.610947100	8.8.8.8	192.168.1.4	DNS	221	Standard query response 0x9a15 A browser.events.data.msn.com CNAME global.asimov.events.data.trafficmanager.net
59	6.610947100	8.8.8.8	192.168.1.4	DNS	179	Standard query response 0x9a34 HTTPS edge.microsoft.com CNAME edge.microsoft.com.ax-0002.ax-msedge.net SOA
60	6.610947100	8.8.8.8	192.168.1.4	DNS	178	Standard query response 0x3557 A edge.microsoft.com CNAME edge.microsoft.com.ax-0002.ax-msedge.net CNAME ax
62	6.628307800	192.168.1.4	8.8.8.8	DNS	69	Standard query 0x17a6 A c.msn.com
63	6.629051400	192.168.1.4	8.8.8.8	DNS	69	Standard query 0x17a6 A c.msn.com
64	6.630122600	192.168.1.4	8.8.8.8	DNS	84	Standard query 0xfab3 HTTPS sb.scorecardresearch.com
65	6.630827200	8.8.8.8	192.168.1.4	DNS	308	Standard query response 0x7e45 HTTPS c.msn.com CNAME c-msn-afd.trafficmanager.net CNAME idsyncprod-cehbcsbu
66	6.631792000	8.8.8.8	192.168.1.4	DNS	297	Standard query response 0x17a6 A c.msn.com CNAME c-msn-afd.trafficmanager.net CNAME idsyncprod-cehbcsbu
67	6.631843900	192.168.1.4	8.8.8.8	DNS	84	Standard query 0x3edf A sb.scorecardresearch.com
68	6.632061300	8.8.8.8	192.168.1.4	DNS	84	Standard query response 0xfab3 HTTPS sb.scorecardresearch.com
70	6.634778800	8.8.8.8	192.168.1.4	DNS	84	Standard query response 0x3edf A sb.scorecardresearch.com
71	6.640427400	192.168.1.4	8.8.8.8	DNS	84	Standard query 0xaa26 A sb.scorecardresearch.com
72	6.640660900	8.8.8.8	192.168.1.4	DNS	84	Standard query response 0xaa26 A sb.scorecardresearch.com
73	6.651251500	192.168.1.4	8.8.8.8	DNS	84	Standard query 0xb044 HTTPS sb.scorecardresearch.com
74	6.653032200	192.168.1.4	8.8.8.8	DNS	84	Standard query 0xb745 A sb.scorecardresearch.com
75	6.656839700	8.8.8.8	192.168.1.4	DNS	84	Standard query response 0xb044 HTTPS sb.scorecardresearch.com
76	6.658187500	8.8.8.8	192.168.1.4	DNS	84	Standard query response 0xb745 A sb.scorecardresearch.com
100	6.823844700	192.168.1.4	8.8.8.8	DNS	74	Standard query 0x7581 HTTPS assets.msn.com
101	6.824449400	192.168.1.4	8.8.8.8	DNS	74	Standard query 0x23d8 A assets.msn.com
108	6.878431300	8.8.8.8	192.168.1.4	DNS	275	Standard query response 0x7581 HTTPS assets.msn.com CNAME assets-msn-com-world-atm-default.trafficmanager.net

Frame 52: Packet, 87 bytes on wire (696 bits), 87 bytes captured (696 bits) on interface \Device\NPF{...}

Ethernet II, Src: Intel_00:0c:29:14:00:00 (18:0e:0f:a0:ec:a4), Dst: 74:ed:42:81:2d:a0 (74:ed:42:81:2d:a0)

Internet Protocol Version 4, Src: 192.168.1.4, Dst: 8.8.8.8

User Datagram Protocol, Src Port: 63611, Dst Port: 53

Domain Name System (query)

0000 74 ed 42 81 2d a0 18 5e 0f a0 ec a4 08 00 45 00 t BE:
0010 00 49 3c cb 00 00 00 11 2c 1d c9 a0 01 04 00 ..I.....
0020 00 08 f6 7b 00 35 00 35 8e f1 cb ba 01 00 00 ...{ 5 5
0030 00 00 00 00 00 00 07 62 72 6f 77 73 65 72 06 65b rowser-e
0040 76 65 6e 74 73 04 64 61 74 61 03 6d 73 6e 03 63vents da ta msn c
0050 6f 6d 00 00 41 00 01 om : A ..

Explanation:

DNS query is sent to resolve a domain name into an IP address before establishing communication.

4.2 TCP Packet Analysis

Protocol: TCP

Purpose: Reliable data transmission

Observed:

- 3-way handshake process:
 1. SYN
 2. SYN-ACK

3. ACK

No.	Time	Source	Destination	Protocol	Length	Info
5	4.776996900	192.168.1.4	172.64.148.235	TLSv1.2	143	Application Data
6	4.807146200	172.64.148.235	192.168.1.4	TCP	58	443 → 59395 [ACK] Seq=1 Ack=90 Win=19 Len=0
7	4.882427600	172.64.148.235	192.168.1.4	TLSv1.2	136	Application Data
23	4.975642000	104.26.13.20	192.168.1.4	TCP	54	443 → 59404 [ACK] Seq=1 Ack=1 Win=16 Len=0
24	4.975703200	192.168.1.4	104.26.13.20	TCP	54	[TCP ACKed unseen segment] 59404 → 443 [ACK] Seq=1 Ack=2 Win=509 Len=0
36	5.107095900	192.168.1.4	172.64.148.235	TCP	54	59395 → 443 [ACK] Seq=90 Ack=79 Win=509 Len=0
37	5.126875000	172.64.148.235	192.168.1.4	TCP	136	[TCP Spurious Retransmission] 443 → 59395 [PSH, ACK] Seq=1 Ack=90 Win=19 Len=78
38	5.126940500	192.168.1.4	172.64.148.235	TCP	66	[TCP Dup ACK 36#1] 59395 → 443 [ACK] Seq=90 Ack=79 Win=509 Len=0 SLE=1 SRE=79
60	6.619318400	192.168.1.4	40.79.150.123	TCP	66	59636 → 443 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM
61	6.623280100	192.168.1.4	150.171.28.11	TCP	66	59637 → 443 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM
69	6.634337800	192.168.1.4	13.107.246.63	TCP	66	59638 → 443 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM
77	6.671513800	150.171.28.11	192.168.1.4	TCP	66	443 → 59637 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1412 WS=256 SACK_PERM
78	6.671689700	192.168.1.4	150.171.28.11	TCP	54	59637 → 443 [ACK] Seq=1 Ack=1 Win=131072 Len=0
79	6.674179600	192.168.1.4	150.171.28.11	TCP	1466	59637 → 443 [ACK] Seq=1 Ack=1 Win=131072 Len=1412 [TCP PDU reassembled in 80]
80	6.674179600	192.168.1.4	150.171.28.11	TLSv1.2	646	Client Hello (SNI=edge.microsoft.com)
81	6.686217100	13.107.246.63	192.168.1.4	TCP	66	443 → 59638 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1412 SACK_PERM WS=512
82	6.686327600	192.168.1.4	13.107.246.63	TCP	54	59638 → 443 [ACK] Seq=1 Ack=1 Win=131072 Len=0
83	6.689858000	192.168.1.4	13.107.246.63	TCP	1466	59638 → 443 [ACK] Seq=1 Ack=1 Win=131072 Len=1412 [TCP PDU reassembled in 84]
84	6.689858000	192.168.1.4	13.107.246.63	TLSv1.3	760	Client Hello (SNI=c.msn.com)
85	6.773271200	150.171.28.11	192.168.1.4	TCP	54	443 → 59637 [ACK] Seq=1 Ack=2005 Win=12583680 Len=0
86	6.773271200	150.171.28.11	192.168.1.4	TLSv1.2	208	Server Hello, Change Cipher Spec, Encrypted Handshake Message
87	6.773271200	13.107.246.63	192.168.1.4	TCP	54	443 → 59638 [ACK] Seq=1 Ack=1413 Win=68608 Len=0
88	6.773271200	13.107.246.63	192.168.1.4	TCP	54	443 → 59638 [ACK] Seq=1 Ack=2119 Win=71680 Len=0
89	6.773271200	13.107.246.63	192.168.1.4	TLSv1.3	153	Hello Retry Request, Change Cipher Spec
90	6.773271200	40.79.150.123	192.168.1.4	TCP	66	443 → 59636 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1412 WS=256 SACK_PERM

> Frame 83: Packet, 1466 bytes on wire (11728 bits), 1466 bytes captured (11728 bits) on interface
> Ethernet II, Src: Intel_80:ec:a4 (18:5e:0f:a0:ec:a4), Dst: 74:ed:42:81:2d:a0 (74:ed:42:81:2d:a0)
> Internet Protocol Version 4, Src: 192.168.1.4, Dst: 13.107.246.63
> Transmission Control Protocol, Src Port: 59638, Dst Port: 443, Seq: 1, Ack: 1, Len: 1412

0000 74 ed 42 81 2d a0 18 5e 0f a0 ec a4 08 00 45 00 t:8---^-----E-
0010 05 ac 3b 04 40 00 80 06 f4 f0 c0 a8 01 04 0d 6b -:;@---k
0020 f6 3f e8 f6 01 bb 34 a0 fa b7 26 7c cb 09 50 10 -:?---4}--&]--P-
0030 02 00 a4 50 00 00 16 03 01 08 41 01 00 08 3d 03 ---P-----A-----
0040 03 32 96 ec 82 f7 e8 0d 39 16 5b 70 be 11 e7 94 -2-----9-[p-----
0050 db fc 34 c2 b5 a3 51 06 cc e1 b1 1c 30 82 41 7d -4---Qf---0-A}
0060 05 20 4d 92 2e 56 14 17 0e 33 99 29 0e c9 d2 e9 -N-V---3-)-----
0070 21 31 e5 ad ef 9a b1 a9 31 76 a8 9e 9f fe a3 fa 11-o---1v-----
0080 5a 3c 00 20 4a 4a 13 01 13 02 13 03 c0 2b c0 2f Zc-3J---+/-

Explanation:

TCP establishes a connection before sending data to ensure reliability.

5. Packet Structure Explanation

Each captured packet contains:

1. Frame Layer
2. Ethernet Layer
3. IP Layer
4. Transport Layer (TCP/UDP)
5. Application Layer (HTTP/DNS)

Wireshark displays packet details in three panels:

- Packet List Pane
- Packet Details Pane
- Packet Bytes Pane

6. Security Observations

During analysis:

- No suspicious packets were detected
- Traffic mainly consisted of DNS, TCP, and HTTPS
- Encrypted traffic prevents viewing sensitive data

Wireshark helps detect:

- Malware traffic
- Unauthorized access
- Network scanning
- Data leakage

7. Advantages of Wireshark

- Free and open-source
- Deep packet inspection
- Supports multiple protocols
- Useful for troubleshooting
- Helps in cybersecurity analysis

8. Conclusion

In this experiment, Wireshark was successfully used to capture and analyze network packets. Different protocols such as DNS, TCP, ICMP, and HTTP were observed and studied.

The analysis helped in understanding:

- How devices communicate over a network
- How domain names are resolved
- How TCP establishes connections
- How encrypted traffic protects data

Wireshark is a powerful tool for network monitoring, troubleshooting, and cybersecurity investigations